Worksheet 8.8

Forces, energy and motion: Summary

Answers

Complete the statements below to compile a summary of this unit. The missing words can be found in the word list below.

1. Speed is a measure of the …rate…… at which an object moves over a distance.
2. …Average… speed can be calculated by dividing the distance travelled by the time taken.
3. …Velocity…. is a measure of the rate of change in position. In order to describe the velocity fully, the …direction… of the change in position must be stated.
4. The …instantaneous…… speed of an object is its speed at a particular instant of time.
5. The …acceleration…. of an object moving in a straight line is a measure of the rate at which it changes speed.
6. The average acceleration of an object can be calculated by …dividing… the change in its speed by the time taken for the change.
7. The force of gravitational attraction to a large object like a planet is called …weight…. .
8. An object will remain at rest, or will not change its speed or direction, unless it is acted upon by an outside, …unbalanced…. force.
9. The acceleration of an object depends on the mass of the object and the …total…. force acting on it.
10. When an object applies a force to a second object, the second object applies an …equal….. and …opposite… force to the first object.
11. …Work….. is done whenever a force is applied to an object and the object moves in the direction of the force.
12. …Energy… can be transferred to an object by doing work on it. Doing work on an object can also convert the energy an object possesses from one form into another.
13. All stored energy is called …potential… energy. …Gravitational…. potential energy and …elastic…. potential energy are examples of stored energy.
14. The Law of Conservation of Energy states that energy is never created or …destroyed… .

average	destroyed	potential	weight
energy	equal	total	opposite
direction	dividing	work	gravitational
unbalanced	instantaneous	acceleration	
elastic	velocity	rate	

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